

Differentially Private Preference Data Synthesis for Large Language Model Alignment

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Abstract

Preference alignment is a central post-training mechanism for large language models (LLMs), yet its reliance on real human preference data creates a direct privacy challenge: preference datasets often contain sensitive user prompts, model responses, and human judgments. We introduce **DPPrefSyn**, a differentially private (DP) algorithm for synthesizing preference data that supports privacy-preserving LLM alignment. DPPrefSyn is grounded in the Bradley–Terry preference model and in the geometric structure of pairwise preference representations. It privately learns preference structure from the sensitive dataset and then combines the learned models with public prompts to construct high-quality synthetic preference pairs. The algorithm captures heterogeneous human feedback through shared linear reward structure within preference clusters and uses DP Principal Component Analysis (DP-PCA) to improve the accuracy and stability of private learning. Across question answering and summarization benchmarks, DPPrefSyn delivers competitive alignment utility under strong DP guarantees. These results establish DP synthetic preference data as a practical interface between privacy protection and modern preference alignment. To the best of our knowledge, this is the first work to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

1. Introduction

Preference alignment methods, including RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024), have become a standard mechanism for adapting LLMs to human expectations. These methods operate on pairwise preference data:

annotators compare two candidate responses to a prompt and identify the response that better satisfies the intended criterion. The resulting comparisons provide a training signal that encourages the model to rank preferred responses above less preferred alternatives (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). Because this paradigm has proved effective for improving instruction following and response quality, it now supports a wide range of LLM applications, including chat assistants (Achiam et al., 2023), mathematical reasoning systems (Shao et al., 2024), and code generation models (Shen et al., 2023).

The same data that makes preference alignment effective also makes it privacy-sensitive. Real user prompts can reveal health information, identity attributes, or other personal content, while preference annotations may disclose private beliefs, values, or behavioral regularities (Li et al., 2023; Yu et al., 2024). Differential privacy (DP) (Dwork et al., 2006b) offers a rigorous framework for mitigating these risks, and prior work has produced encouraging privacy-preserving alignment methods (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a). Nevertheless, existing approaches typically protect only a subset of the preference data, such as user prompts (Yu et al., 2024) or preference labels (Chowdhury et al., 2024; Zhang et al., 2025), leaving the remaining prompts, responses, or annotations available as potential sources of leakage. They are also often tied to a specific post-training procedure, such as RLHF (Wu et al., 2023a), rather than producing a reusable private data resource compatible with newer alignment methods such as DPO. Finally, because human preference annotations are expensive and difficult to scale, methods that train directly on limited private preference datasets can struggle to deliver high-quality alignment. These limitations motivate the central question of this work:

Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining compatibility with emerging alignment methodologies?

This paper answers that question affirmatively by developing a DP synthetic-data route to preference alignment. The contribution is not merely a private fine-tuning procedure, but a reusable mechanism for converting sensitive prefer-

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ence data into synthetic preference pairs that can support multiple downstream alignment methods and model choices. Our main contributions are as follows:

- We introduce DPPrefSyn (Figure 1), an algorithm for generating DP synthetic preference data for LLM alignment. DPPrefSyn builds on the Bradley–Terry preference model and the geometry of pairwise preference embeddings. Its design has three coordinated components: 1) *Capturing diverse human preferences via shared linear structure of rewards and clustering*. The Bradley–Terry model with a linear reward parameterization allows preference samples to be grouped by embedding differences between concatenated prompt–preferred and prompt–dispreferred pairs, so that each cluster can be represented by a shared reward function. 2) *Improving learning efficiency through DP-PCA*. We apply DP-PCA to project high-dimensional embeddings into a compact private subspace, improving sample efficiency and clustering stability while preserving preference-relevant signal. 3) *Saving privacy budgets by using public prompts*. We generate candidate responses from public prompts and use private reward models to select preference pairs, concentrating the privacy budget on private preference modeling rather than prompt synthesis.
- We provide a formal privacy analysis showing that DPPrefSyn satisfies (ϵ, δ) -DP. The analysis allocates ϵ_0 to DP-PCA (Amin et al., 2019), ϵ_1 to DP-KMeans (Su et al., 2016), and the remaining privacy budget to per-cluster reward-model training with DP-SGD. We compose the DP-SGD cost using the Privacy Random Variable (PRV) accountant (Gopi et al., 2021). By the post-processing property of DP, the generated synthetic preference dataset can then be reused across alignment algorithms and LLMs without additional privacy expenditure.
- We evaluate DPPrefSyn on standard preference-alignment benchmarks, including OpenAssistant question answering (Köpf et al., 2023), Anthropic-HH question answering (Bai et al., 2022), and TL;DR summarization (Stiennon et al., 2020). The experiments show that DPPrefSyn combines strong privacy guarantees with competitive utility across tasks. For example, at $\epsilon = 2$, DPO fine-tuning of Pythia-2.8B on DPPrefSyn data achieves a GPT-4o win rate of 56.48% on Anthropic-HH, compared with 37.02% for fine-tuning on real data. The same synthetic-data interface remains applicable to multiple preference optimization methods, including DPO and RLHF, addressing the method-specific nature of prior private post-training approaches.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al. (2023a) introduce a DP framework for reinforcement-

learning-based LLM alignment by adapting PPO to the DP setting. Their formulation provides an important private RLHF baseline, but it is *limited to PPO* even as preference alignment increasingly relies on alternative objectives. Chowdhury et al. (2024) study reward estimation from preference-based feedback under label-DP (Ghazi et al., 2021), and Zhang et al. (2025) propose AUP-RLHF as a user-level label-DP framework for RLHF. These methods protect preference labels, but they *do not address the sensitivity of prompts or responses themselves*. They also remain *constrained by the limited private user preference data* available for alignment, a substantive limitation because human preference annotations are costly to obtain at scale. Yu et al. (2024) instead protect sensitive user *instructions* by generating DP synthetic *instructions* to replace real instructions during data annotation and model fine-tuning.

Beyond preference alignment, a substantial line of work studies private LLM fine-tuning with general labeled data rather than pairwise preference comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022). A complementary set of recent methods protects in-context prompts for in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b). These settings share the goal of privacy-preserving LLM adaptation, but they do not directly solve the problem of producing reusable private preference data for alignment.

Differentially Private Synthetic Text Generation. DPPrefSyn is situated within DP synthetic text generation, where two representative strategies have emerged. The first privately fine-tunes a language model on sensitive data and then uses the resulting model to generate synthetic text (Yue et al., 2023; Mattern et al., 2022; Miresghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For example, Yu et al. (2024) privately fine-tune LLaMA on private instruction data and then use the resulting model to synthesize text for supervised instruction tuning.

Because DP fine-tuning can be expensive and, for non-public language models, sometimes infeasible, a second line of work generates DP synthetic data by prompting LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). These approaches typically sample candidate outputs through API access and then select outputs that are close to private data under a privacy-preserving criterion. For instance, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples using embedding-space similarity to private data. *Our work advances this API-based synthetic-data direction from text generation to pairwise preference synthesis for LLM alignment.*

Diversity in Human Preferences. Human feedback is not

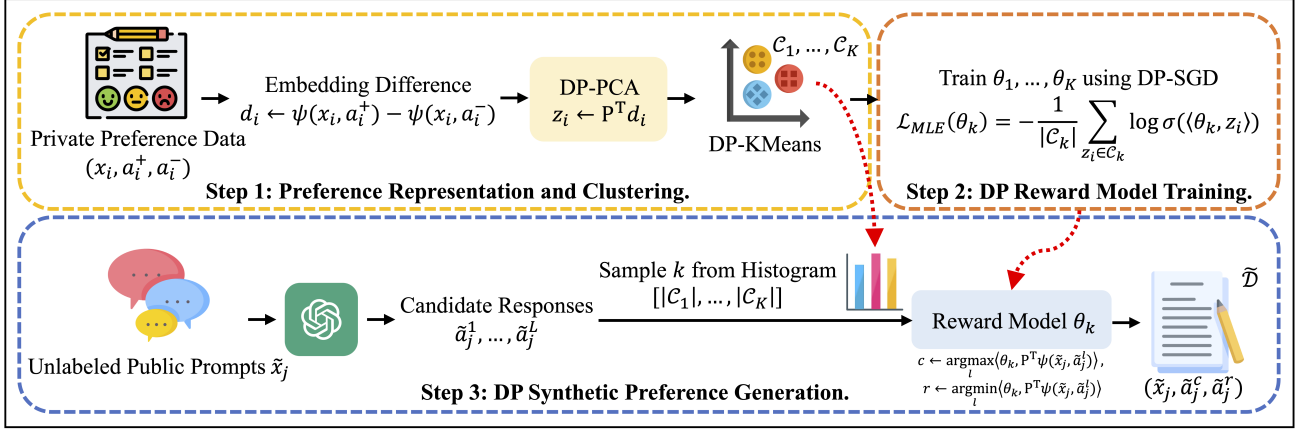


Figure 1. Overview of DPPrefSyn. The algorithm synthesizes DP preference data in three stages: 1) representing preference samples as embedding differences and clustering them with DP-PCA and DP-KMeans, 2) training cluster-specific DP reward models with DP-SGD, and 3) generating synthetic preference pairs from public prompts using the DP-protected distribution over learned reward functions.

a homogeneous signal: recent work emphasizes substantial diversity in preferences (Denton et al., 2021; Aroyo et al., 2023a;b; Chakraborty et al., 2024). Sources of variation include sociodemographic differences (e.g., race, gender, age), personal beliefs and biases, levels of domain expertise, and ambiguity in natural language (Sandri et al., 2023; Vogels, 2021). This observation has motivated methods for personalized preference alignment that better reflect the values of different user groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). For example, Singh et al. (2025) propose a meta-learning framework in which an LLM adapts to individual users from a small number of preference examples. DPPrefSyn incorporates this heterogeneity into private synthesis by generating DP synthetic preference data that captures the underlying diversity of human feedback, which improves alignment performance relative to designs that collapse all preferences into a single cluster (as shown in Section 5.3).

3. Problem Definition

Preference Alignment of LLMs. We study LLM preference alignment from pairwise human feedback. The available preference dataset is $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$, where each record contains a prompt x_i and two candidate responses a_i^+, a_i^- . The notation indicates that human annotators prefer a_i^+ to a_i^- for the corresponding prompt.

Following prior works on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we model human preferences using the Bradley–Terry model (Bradley & Terry, 1952):

$$\begin{aligned} \mathbb{P}[a_i^+ \succ a_i^- | x_i] &= \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} \\ &= \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)), \end{aligned}$$

where $a_i^+ \succ a_i^-$ denotes the preference of a_i^+ over a_i^- , $r^*(x, a)$ is the latent reward function, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function.

This probabilistic formulation underlies both RLHF, which explicitly estimates a reward model from preference comparisons, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy.

Differentially Private Preference Data Synthesis. The private dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ contains prompts, responses, and preference labels that may reveal individual-level information. We seek to protect this dataset against an adversary who observes the outputs of a fine-tuned LLM and attempts to infer private information about any individual prompt, response, or label. DPPrefSyn addresses this objective by generating a DP synthetic preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that retains the preference information needed for alignment while satisfying a formal privacy guarantee. By the post-processing property of DP, $\tilde{\mathcal{D}}$ can subsequently be reused with different preference alignment methods and LLMs without incurring additional privacy cost.

We use the standard definition of (ϵ, δ) -differential privacy.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure. Human preferences in $\mathcal{D}_{\text{priv}}$ are generally not homogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Different annotators may assign different weight to factual accuracy, politeness, creativity, clarity, or other response

attributes, producing heterogeneous and sometimes conflicting preference signals. Effective synthesis of $\tilde{\mathcal{D}}$ therefore requires a representation that preserves preference diversity rather than averaging it away.

To model this structure, we assume a linear form for the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we adopt a linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where $\phi(x, a) \in \mathbb{R}^d$ is the *shared* feature representation and $\theta \in \mathbb{R}^d$ is a *user-dependent* parameter vector.

This parameterization provides the expressiveness needed to distinguish preference styles while retaining the tractability required under limited data and DP constraints. In particular, users or preference clusters may share a semantic feature space (i.e., $\phi(x, a)$) while differing in the weights assigned to those features (i.e., θ). The resulting model captures structured heterogeneity without requiring a separately fine-tuned neural reward model for each preference pattern. As shown empirically in Appendix B.6, this linear reward structure captures preference signals effectively and achieves competitive performance relative to a fully fine-tuned GPT-2 reward model, despite its simplicity.

4. Proposed Method

4.1. Key Components of the Algorithm Design

We first isolate the design principles that make DP preference synthesis feasible: preference heterogeneity must be represented explicitly, high-dimensional embeddings must be made learnable under privacy constraints, and the privacy budget should be spent on sensitive preference information rather than on nonsensitive prompt generation.

Leveraging Linear Reward Structure for Preference Clustering. The linear reward model indicates that

$$\mathbb{P}[a_i^+ \succ a_i^- | x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

For a tuple (x_i, a_i^+, a_i^-) , the event $a_i^+ \succ a_i^-$ therefore implies that $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$; equivalently, the preference vector θ is strongly aligned with the feature difference $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. Annotators or preference regimes with similar reward weights should consequently induce feature-difference vectors that point in similar directions in the embedding space. DPPrefSyn uses this structure to cluster preference samples by the difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$, so that samples within a cluster can be approximated by a shared reward function parameterized by a common preference vector θ . Appendix B.9 confirms that the resulting clusters are meaningful and interpretable in terms of user preference styles.

Adopting DP-PCA to Obtain Effective Features. A natural instantiation of $\phi(x, a)$ is the embedding of the corresponding (prompt, response) pair, because such embeddings encode semantic and stylistic distinctions that are relevant to human preference judgments. These embeddings are typically high-dimensional (e.g., 1024 dimensions), which makes private clustering and per-cluster reward estimation sample-inefficient. The issue is especially acute after clustering, where each reward model is trained on only a subset of the private data. DPPrefSyn therefore applies differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to obtain a compact private feature representation for both clustering and reward-model training. This dimensionality reduction improves sample efficiency and clustering stability while retaining the preference signal needed for downstream alignment.

Utilizing Public Prompts to Save DP Budget. Jointly synthesizing prompts, preferred responses, and less preferred responses would spend privacy budget on every component of the preference tuple and can substantially degrade utility. DPPrefSyn instead uses public prompts, which allows the private computation to focus on modeling preferences over responses. For each public prompt, an LLM generates multiple candidate responses; the DP-trained distribution of reward functions then selects preferred and less preferred responses to form synthetic pairs. Although public prompts may not match the private prompt distribution exactly, the experiments show that the resulting synthetic preference data preserves useful statistical preference structure and remains effective for downstream alignment.

4.2. DPPrefSyn Algorithm

Algorithm 1 formalizes DPPrefSyn, which transforms the private preference dataset $\mathcal{D}_{\text{priv}}$ into a DP synthetic preference dataset through three stages.

Step 1: Preference Representation and Clustering. DPPrefSyn represents each private comparison through the difference between two prompt-response embeddings. For a sample (x_i, a_i^+, a_i^-) , we concatenate the prompt with each response to form $[x_i; a_i^+]$ and $[x_i; a_i^-]$, embed both texts with the public sentence embedding model ψ , and define the preference difference as $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3).

The high-dimensional difference vectors are then reduced privately by applying DP-PCA (Amin et al., 2019) to $\{d_i\}_{i=1}^n$, which returns a projection matrix $\mathbf{P} \in \mathbb{R}^{d \times p}$ under ε_0 -DP (Line 5). DP-PCA (Amin et al., 2019) approximates the covariance eigendecomposition by privately estimating eigenvalues and eigenvectors. Each difference vector is projected as $z_i \leftarrow \mathbf{P}^\top d_i$ (Line 6). DPPrefSyn then applies DP-KMeans clustering (Su et al., 2016) to $\{z_i\}_{i=1}^n$ under ε_1 -DP, producing K preference clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$.

(Line 7). DP-KMeans (Su et al., 2016) supplies the differentially private analogue of Lloyd-style clustering used in this stage.

Step 2: DP Reward Model Training. For each cluster C_k , DPPrefSyn estimates a cluster-specific preference vector $\theta_k \in \mathbb{R}^p$ by minimizing the negative log-likelihood induced by the Bradley–Terry model (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in C_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i is the DP-PCA-projected embedding difference from Step 1. Each cluster reward model is optimized with DP-SGD (Abadi et al., 2016) using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. DPPrefSyn records the private preference mixture through the cluster-size histogram $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ (Line 11) and normalizes it into a sampling distribution $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ over clusters (Line 12).

For each public prompt $\tilde{x}_j$, an LLM is sampled at a relatively high temperature to produce L diverse candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). DPPrefSyn samples a cluster index $k \sim \mathbf{p}$ from the DP histogram (Line 15) and scores the candidate responses with the corresponding reward vector θ_k . Specifically, it embeds each pair $(\tilde{x}_j, \tilde{a}_j^l)$, projects the embedding using $\mathbf{P}$, and assigns reward $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. The highest-scoring response becomes the preferred response and the lowest-scoring response becomes the less preferred response: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16–17). Candidate pairs with a small reward gap (e.g., < 0.5) are filtered out to maintain preference quality; all remaining triples $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ are added to $\tilde{\mathcal{D}}$. Repeating this procedure over public prompts yields the final DP synthetic preference dataset.

Privacy Analysis of DPPrefSyn. The private computation spends ε_0 on DP-PCA (Line 5) and ε_1 on DP-KMeans (Line 7); the remaining privacy budget is assigned to per-cluster reward-model training with DP-SGD and noise multiplier σ_2 (Line 9). Because the clusters are disjoint, parallel composition applies to reward-model training: changing one private sample affects only one cluster, so the relevant DP-SGD cost is bounded through the cluster on which the privacy loss is largest. We discard clusters with too few samples to maintain a known lower bound on cluster size. The Privacy Random Variable (PRV) accountant (Gopi et al., 2021) composes the per-step DP-SGD costs and ensures that this stage satisfies $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. All subsequent operations use public data or DP post-processing and therefore incur no additional privacy cost. Appendix A provides the full proof and privacy-accounting code.

Discussion on Prompt Distribution Shift. DPPrefSyn is

Algorithm 1 DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $C_k$  do
9:   Train a linear reward model  $\theta_k$  on  $C_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
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designed to remain useful when public and private prompt distributions differ. The key observation is that preference diversity is represented by θ_k , the parameter of the linear reward model, rather than solely by $\phi(x, a)$, the feature vector of a particular prompt-response pair. A user may issue many different prompts while maintaining a relatively stable preference style; consequently, preserving the distribution over θ_k can preserve preference patterns even when prompt content shifts. DPPrefSyn implements this idea by sampling a cluster index k from the DP histogram (Line 15) for each public prompt and then selecting response pairs according to the corresponding reward model. Thus, although private and public prompts need not be identically distributed, the synthetic data can still reflect the preference structure learned from the private dataset.

5. Experiments

Datasets. We evaluate DPPrefSyn on preference-alignment tasks spanning question answering and summarization. For question answering, we use OpenAssistant (Köpf et al., 2023), which contains assistant-style conversations, and Anthropic-HH (Bai et al., 2022), which provides human preference comparisons emphasizing helpfulness and harmlessness. For summarization, we use TL;DR (Stiennon et al., 2020), which contains human preferences over summary pairs. Public prompts are drawn from Alpaca (Taori et al.,

2023) for OpenAssistant QA, SafeRLHF (Ji et al., 2024) for Anthropic-HH QA, and XSum (Narayan et al., 2018) for TL;DR summarization. Appendix B.2 provides dataset details.

Setup for Algorithm 1. Unless otherwise specified, DP-PrefSyn uses BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ , projected dimension $p = 20$ for DP-PCA, and $K = 5$ clusters for DP-KMeans. DP-SGD uses learning rate 0.1, batch size 4, 4 training epochs, and gradient clipping norm 1.0. Candidate responses are sampled from the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) at temperature 0.9, with $L = 5$ candidates per prompt. We evaluate privacy budgets $\varepsilon = 0.5, 1, 2, 4, 8$ and set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We study different generator models in Table 3, and embedding models and hyperparameters in Section 5.3.

Evaluation. The primary downstream model is Pythia-2.8B (Biderman et al., 2023). We first perform supervised fine-tuning (SFT), using the preferred response in each preference pair as the training target, and then apply DPO (Rafailov et al., 2024) to further optimize the SFT model on preference pairs. Following Rafailov et al. (2024), we report GPT-4o win rates by comparing model-generated responses with the preferred responses in the test set. *Additional downstream models and alignment methods are evaluated in Section 5.2.*

Baselines. The main baseline is DP-FT, which fine-tunes the downstream model directly on real preference data with DP-SGD (Abadi et al., 2016). This comparison distinguishes a synthetic-data privacy interface from a model-specific private training procedure: DPPrefSyn produces a DP synthetic dataset that can be reused across models and alignment algorithms without additional privacy cost, whereas DP-FT spends the privacy budget to train one privatized model. We also include a fully private $\varepsilon = 0$ baseline, corresponding to the base model without preference fine-tuning.

Details about the setups, hyperparameters, metrics, and baselines are provided in Appendix B.3.

5.1. Understanding the Performance of DPPrefSyn

We first evaluate the central privacy-utility claim: whether a reusable DP synthetic preference dataset can outperform direct private fine-tuning while preserving formal privacy. Table 1 reports the main results.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? DPPrefSyn consistently delivers a stronger privacy-utility trade-off than DP-FT across all three tasks and across $\varepsilon = 0.5, 1, 2, 4, 8$; in these evaluations, it also exceeds the utility of direct fine-tuning on real private data without DP constraints (DP-FT with $\varepsilon = \infty$). On OpenAssistant QA at $\varepsilon = 2$, DPPrefSyn

reaches 9.10% after SFT and 11.04% after DPO, whereas DP-FT reaches 4.38% after SFT and 6.18% after DPO. DP-PrefSyn also surpasses the non-private DP-FT row with $\varepsilon = \infty$, which obtains 4.75% after SFT and 8.20% after DPO. The same pattern appears on Anthropic-HH at $\varepsilon = 2$: DPPrefSyn obtains 55.21% after SFT and 56.48% after DPO, compared with 31.53% and 37.02% for DP-FT. It also exceeds direct fine-tuning on real data without DP constraints, where DP-FT with $\varepsilon = \infty$ achieves 31.98% after SFT and 38.72% after DPO.

These results reveal two complementary advantages of DP synthetic preference data. **First**, DPPrefSyn substantially improves SFT utility relative to directly applying SFT to private data. On TL;DR summarization at $\varepsilon = 2$, DPPrefSyn achieves 62.49% after SFT, compared with 22.57% for SFT on private data. This gain is consistent with the construction of DPPrefSyn: candidate responses are generated by a high-performance LLM and then filtered through private reward functions, yielding effective supervised targets. The advantage persists when public prompts are limited. On OpenAssistant QA at $\varepsilon = 4$ with only 5K public prompts, DPPrefSyn obtains 6.80% after SFT, compared with 4.46% for SFT directly on the approximately 14K-sample private dataset (Table 4). **Second**, DPO applied after SFT further improves performance on DPPrefSyn data, indicating that the synthetic preference pairs encode preference distinctions that remain useful for pairwise optimization. The incremental DPO gain is smaller than under direct private-data fine-tuning, plausibly because the SFT model trained on DPPrefSyn data is already stronger; nevertheless, the full SFT+DPO pipeline with DP synthetic data produces larger overall gains than the corresponding pipeline on real private data.

RQ2: How does DPPrefSyn perform across different privacy budget ε ? First, Table 1 shows that DPPrefSyn maintains its advantage over DP-FT as the privacy budget varies across $\varepsilon = 0.5, 1, 2, 4, 8$ and ∞ , demonstrating favorable scaling with the privacy budget. **Second**, DPPrefSyn can be less sensitive to DP noise than DP-FT in some settings. On OpenAssistant with Llama-3-8B as the downstream model, DP-FT DPO performance decreases from 59.13% to 45.79% when ε decreases from ∞ to 4, whereas DPPrefSyn decreases only from 61.52% to 60.25% (see Table 3 in Section 5.2). This pattern is consistent with the different locations of privacy noise: DP-FT perturbs model-parameter updates through DP-SGD, whereas DPPrefSyn injects privacy noise during data synthesis before downstream alignment.

5.2. Understanding the Properties of DPPrefSyn

We next examine whether the synthetic-data interface remains useful beyond the default Pythia-2.8B DPO setting. The analyses cover alternative alignment methods, down-

Table 1. GPT-4o win rates on test sets for Pythia-2.8B after SFT and DPO. $\varepsilon = 0$ denotes the base, fully private, non-finetuned LLM. Across $\varepsilon = 0.5, 1, 2, 4, 8$, DPPrefSyn obtains higher win rates than DP-FT and also exceeds DP-FT with $\varepsilon = \infty$ in these evaluations. Results report mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT SFT+DPO	3.96 _{0.20} 5.65 _{0.32}	4.29 _{0.21} 6.04 _{0.58}	4.38 _{0.23} 6.18 _{0.83}	4.46 _{0.18} 6.12 _{0.63}	4.49 _{0.30} 6.32 _{0.71}	4.75 _{0.27} 8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT SFT+DPO	8.96 _{0.39} 9.86_{0.59}	9.07 _{0.32} 10.31_{0.54}	9.10 _{0.51} 11.04_{0.92}	9.33 _{0.56} 11.85_{0.99}	9.63 _{0.38} 12.05_{0.82}	9.80 _{0.58} 12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT SFT+DPO	29.77 _{0.76} 35.00 _{0.49}	31.10 _{0.72} 36.27 _{0.63}	31.53 _{0.77} 37.02 _{0.72}	31.29 _{0.65} 36.74 _{0.90}	31.37 _{0.67} 36.94 _{1.06}	31.98 _{0.56} 38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT SFT+DPO	54.90 _{0.66} 55.08_{1.04}	55.19 _{0.91} 55.96_{0.80}	55.21 _{0.82} 56.48_{0.61}	54.95 _{1.10} 56.51_{1.93}	55.63 _{0.86} 56.86_{1.17}	55.95 _{0.70} 57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT SFT+DPO	21.19 _{1.12} 59.69 _{0.82}	22.41 _{1.09} 62.21 _{1.12}	22.57 _{0.65} 62.02 _{1.67}	22.45 _{0.90} 63.00 _{0.73}	23.04 _{0.79} 63.06 _{0.84}	23.82 _{1.26} 64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT SFT+DPO	61.92 _{1.06} 67.50_{0.79}	62.14 _{1.61} 68.07_{1.95}	62.49 _{1.48} 68.54_{2.76}	62.72 _{1.66} 68.56_{1.59}	62.29 _{1.14} 71.01_{2.17}	62.55 _{1.23} 72.20_{1.08}

Table 2. DPPrefSyn achieves higher win rates than fine-tuning on private data across SFT, DPO, and PPO, while providing strong privacy guarantees on OpenAssistant. Results show mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

Table 3. DPPrefSyn achieves higher win rates than fine-tuning on private data across different downstream LLMs after DPO, while providing strong privacy guarantees on OpenAssistant. DPPrefSyn can also benefit from more powerful generator LLMs.

Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
		DPPrefSyn	Qwen-3-4B-Instruct Llama-3-8B-Instruct	86.52 51.69	87.08 52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
		DPPrefSyn	Qwen-3-4B-Instruct Llama-3-8B-Instruct	36.24 36.01	37.08 36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
		DPPrefSyn	Qwen-3-4B-Instruct Llama-3-8B-Instruct	60.25 42.28	61.52 44.10

stream LLM families, public-data scale, and empirical membership inference risk.

RQ3: Is DPPrefSyn effective across different preference alignment methods? DPPrefSyn functions as a reusable private data layer for multiple alignment objectives. Table 2 evaluates SFT, DPO, and PPO. For PPO, we train K reward models, one per cluster, to represent diverse preference patterns; PPO then optimizes the policy using the average output of these K reward models as the reward. Across SFT, DPO, and PPO, DPPrefSyn provides higher utility than fine-tuning on private data while preserving strong privacy guarantees.

RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? DPPrefSyn remains effective across multiple downstream LLM families. Table 3 evaluates Qwen-3-4B, Gemma-2-2B, and

Table 4. DPPrefSyn achieves higher win rates than fine-tuning on private data with varying public data size m . Results show mean and standard deviation over 5 random seeds.

	DPPrefSyn ($\varepsilon = 4$)			DP-FT ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
Data Type (Size)	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}

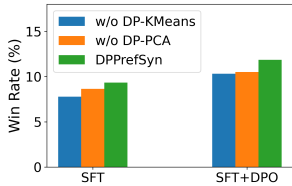
Llama-3-8B as downstream models, with Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generator models. After DPO, DPPrefSyn outperforms fine-tuning on real data while preserving privacy across these model families. For example, with Qwen3-4B-Instruct-2507 as the generator, DPPrefSyn reaches an 86.52% win rate on Qwen-3-4B at $\varepsilon = 4$, compared with 59.41% for non-private fine-tuning on real data. The comparison between generators also shows that stronger generation quality improves DPPrefSyn, with Qwen3-4B-Instruct-2507 outperforming Llama-3-8B-Instruct.

RQ5: Can DPPrefSyn be effective with varying scales of public data? Table 4 studies public prompt size m . The private OpenAssistant dataset contains 14K preference annotations, and the full public Alpaca dataset contains 52K prompts. We sample public-prompt subsets of size $m = 5K, 10K, 52K$ and generate DPPrefSyn data from each subset. Even with only 5K public prompts, DPPrefSyn achieves 6.80% after SFT and 9.44% after DPO at $\varepsilon = 4$, exceeding the DP-FT baseline scores of 4.46% and 6.12%. Thus, **DPPrefSyn remains effective even when public prompts are limited.** As m increases, the margin over DP-FT becomes larger, showing that DPPrefSyn can leverage larger pools of public prompts to mitigate scarcity in private preference data.

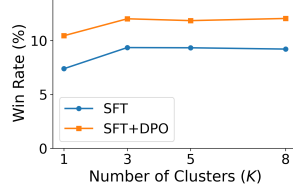
RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? Table 5 reports membership inference attacks (MIAs) (Shokri et al., 2017) against DPO-fine-tuned downstream models on OpenAssistant, using AUC as the

Table 5. Membership inference attack AUC scores (Feng et al., 2025) on OpenAssistant. DPPrefSyn yields lower AUC values than DP-FT, close to random guessing. Results report mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64_{1.29}	49.98_{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51_{1.38}	50.43_{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45_{0.87}	50.67_{0.96}



(a) Ablation removing DP-KMeans or DP-PCA.



(b) Effect of varying the number of clusters.

attack metric. Following prior work (Feng et al., 2025), the attacker attempts to determine whether a private tuple (x, a^+, a^-) participated in preference alignment. DP-FT does not sufficiently mitigate this attack, whereas **DPPrefSyn consistently produces lower AUC scores close to random guessing**. This empirical privacy behavior is consistent with the synthetic-data construction, which reduces direct overfitting to real private examples. Details are provided in Appendix B.4.

5.3. Validating the Design of DPPrefSyn

We validate DPPrefSyn by isolating the algorithmic choices that support its empirical performance.

RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance? We evaluate preference-diversity modeling on OpenAssistant at $\varepsilon = 4$ by removing DP-KMeans or DP-PCA and by varying the number of clusters K . **(i) Removing either DP-KMeans or DP-PCA.** Figure 2a shows that both components contribute to performance. Without DP-KMeans, all samples are forced into one cluster, eliminating explicit preference heterogeneity. Without DP-PCA, clustering and learning must operate in a high-dimensional space, which produces ineffective clustering and sample-inefficient reward estimation. Using multiple clusters together with DP-PCA improves performance over the single-cluster design. **(ii) Varying the number of clusters K .** Figure 2b evaluates $K = 1, 3, 5, 8$. Performance improves when moving to $K = 3$ and then remains relatively stable for $K = 5, 8$, indicating robustness to moderate choices of K . These results support the central design

claim that capturing diverse preferences through multiple clusters improves DP preference synthesis.

RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance? Appendix B.5 ablates the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filtering threshold, and privacy-budget allocation. Across bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), DP-PrefSyn remains effective, although embedding quality affects the achieved win rate. The best overall DPO performance occurs at $p = 20$; $L = 5$ is a strong default; a moderate filtering threshold of 0.5 slightly improves over no filtering; and a threshold of 1 is too aggressive and reduces performance. DPPrefSyn is also robust to reasonable changes in ε_0 and ε_1 .

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? To rule out confounding from prompt or generator quality, we conduct controlled ablations in which all methods share the same prompt set or response generator. Under these controls, DPPrefSyn continues to outperform DP-FT (Appendix B.7), showing that **its performance gains do not merely arise from higher-quality public prompts or stronger response generators**.

6. Conclusion

We introduced DPPrefSyn, a DP algorithm for synthesizing preference data from private pairwise human feedback. By clustering preference representations, estimating cluster-specific linear reward models in a DP-PCA subspace, and using public prompts for candidate generation, DPPrefSyn converts sensitive preference data into a reusable synthetic resource for LLM alignment. The experiments demonstrate that this synthetic-data interface preserves formal privacy, supports multiple alignment methods and downstream models, and outperforms direct private fine-tuning across the reported benchmarks. These findings establish DP synthetic preference data as an effective approach for privacy-preserving LLM alignment under the evaluated settings.

7. Limitations

DPPrefSyn requires LLM API access for candidate response generation, which can introduce additional deployment cost. The present study evaluates English text-based preference data; extending the approach to multilingual and multimodal preference settings remains open. The experiments also cover downstream and generator models up to 8B parameters because of computational constraints.

Impact statement

This work supports preference alignment of large language models while reducing exposure of sensitive preference data. By providing a formal DP route to synthetic preference data, it contributes to methods that balance the benefits of human feedback with the need to protect individual privacy, encourage responsible data use, and strengthen trust in data-driven AI systems.

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A. Privacy Analysis

This section gives the formal privacy analysis for Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We begin with the privacy-curve tools used to compose the private subroutines in DPPrefSyn.

Many DP algorithms, including the Gaussian mechanism, are characterized by a family of (ε, δ) -DP guarantees. For each fixed ε , the mechanism induces a corresponding value $\delta(\varepsilon)$ such that the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Privacy curves are useful because they support tighter composition guarantees than standard advanced composition theorems (Dwork et al., 2014). In particular, numerical composition over privacy curves yields tight accounting for composed mechanisms. Gopi et al. (2021) provide privacy curves for several standard mechanisms, including the Gaussian and subsampled Gaussian mechanisms.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where $\varepsilon_{\text{error}}$ is the additive error in ε , δ_{error} is the additive error in δ , and $\varepsilon_{\text{upper}}$ is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{\text{error}}}{k}\right)\right\}.$$

Gopi et al. (2021) also characterize the privacy loss of a subsampled mechanism from the privacy loss of the original mechanism, which is the accounting primitive used to bound DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We now apply these tools to DPPrefSyn.

Proof. The algorithm accesses the private dataset in three places: DP-PCA, which satisfies ε_0 -DP (Amin et al., 2019) in Line 5; DP-KMeans, which satisfies ε_1 -DP (Su et al., 2016) in Line 7; and DP-SGD with noise multiplier σ_2 in Line 9. In the experiments, ε_0 is allocated to DP-PCA and ε_1 to DP-KMeans. The remaining privacy budget is allocated to DP-SGD and composed with the Privacy Random Variable accountant (Gopi et al., 2021).

Line 7 partitions the low-dimensional embeddings $\{z_i\}_{i=1}^n$ into K disjoint DP clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$. A separate linear reward model is then trained on each cluster with DP-SGD in Line 9. By the parallel composition property of DP, because each private sample belongs to exactly one cluster, the reward-training privacy cost is governed by the cluster-specific DP-SGD

analysis rather than by summing over all clusters. To maintain a known sample-size lower bound for this analysis, clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K+4)$ samples are discarded. DP-SGD uses batch size 4, 4 epochs, and gradient clipping norm 1.0. Its per-iteration privacy loss follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4); Theorem A.3 then composes the DP-SGD iterations (Gopi et al., 2021) so that this stage satisfies $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. Table 7 reports the experimental values of ε_0 , ε_1 , and σ_2 .

The following code implements the privacy accounting used in the experiments.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):

    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )

    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )

    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )

    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments are conducted using NVIDIA H100 GPUs, each with 80 GB of memory. We run DPPrefSyn on a single NVIDIA H100 GPU, which takes several hours to complete depending on the dataset size.

B.2. Datasets

In this section, we describe the datasets used in our experiments.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated, human-annotated assistant-style conversation corpus containing 161,443 messages in 35 languages, with 461,292 quality ratings and over 10,000 fully annotated conversation trees. In our experiments, we use only the English subset, consisting of 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference annotations used to train reward models for RLHF. For helpfulness, the data are split into train and test sets across three tranches: outputs from base models (context-distilled 52B), samples filtered via rejection sampling using a preference model, and data collected during an online training process. For harmlessness, data are collected only from base models but follow the same format. The dataset includes 161K training and 8.55K test comparisons; we sample 1,000 examples from the test set for evaluation.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) contains human preference annotations over pairs of summaries, where annotators are asked to choose the best out of two summaries. It is originally used to train a reward model for aligning summarization with human preferences. The summaries come from the TL;DR dataset, and additional validation data come from CNN articles and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; we sample 979 examples from the validation split as our test set.

- **Alpaca.** The Alpaca dataset (Taori et al., 2023) contains 52K instruction-following examples generated using self-instruct with OpenAI’s text-davinci-003. Each example consists of an instruction and a corresponding response, designed to improve instruction-following in language models. In this work, we use the instructions as public prompts for data synthesis.
- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset containing both performance and safety preferences. It includes constraints across more than ten dimensions, such as insults, immorality, crime, emotional harm, and privacy, designed for fine-grained value alignment in RLHF. In this work, we use the 73.9K prompts from the training split as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) is designed for evaluating abstractive single-document summarization systems. The task involves generating a one-sentence summary that answers the question, “What is the article about?” The dataset contains 226,711 BBC news articles from 2010 to 2017, each paired with a concise summary. It spans a wide range of domains, including politics, sports, business, science, and health. It includes 204,045 training, 11,332 validation, and 11,334 test examples. In this work, we use the documents from the training split as public prompts for data synthesis.

B.3. Implementation Details and Hyperparameter Settings

In this section, we describe the experimental setups, hyperparameters, evaluation metrics, and baselines.

Implementation Details of DPPrefSyn. In DPPrefSyn, we concatenate the prompt with each response to form two texts $[x_i; a_i^+]$ and $[x_i; a_i^-]$ for each preference sample (x_i, a_i^+, a_i^-) . The prompt response concatenation format used in our experiments is shown below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

By default, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We use the DP-PCA implementation from Shoemate et al. (2021), based on the algorithm proposed by Amin et al. (2019). We use the DP-KMeans implementation from Holohan et al. (2019), based on the algorithm proposed by Su et al. (2016). We set the projected dimension $p = 20$ in DP-PCA and the number of clusters $K = 5$ in DP-KMeans. When training the reward models using DP-SGD, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples to ensure a known lower bound on the sample size per model. For DP-SGD, we use the SGD optimizer with a learning rate of 0.1. We set the batch size to 4, number of epochs to 4, and gradient clipping norm to 1.0. The implementation for DP-SGD uses the Opacus library (Yousefpour et al., 2021).

By default, we use the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) to generate candidate responses for each public prompt. We set the number of candidates per prompt to $L = 5$. The generation temperature is set to 0.9. The prompt format used for generating multiple completions is shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:

Privacy Parameters in DPPrefSyn. In our experiments, we set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We provide the values of $|\mathcal{D}_{\text{priv}}|$ in Table 6. We present DP parameters ε_0 , ε_1 and σ_2 in Table 7.

Table 6. Sizes of private datasets used in experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DP parameters ε_0 , ε_1 and σ_2 in DPPrefSyn experiments.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. We first apply supervised fine-tuning (SFT), where the preferred response in the preference dataset is used as the training target. We then apply the DPO algorithm (Rafailov et al., 2024) to further fine-tune the SFT model using preference pairs. All models are trained with bfloat16 precision. Following Rafailov et al. (2024), for SFT we use a learning rate of 5×10^{-7} , training for 1 epoch. For DPO, we use $\beta = 0.1$, a learning rate of 1×10^{-6} , and train for 3 epochs. The batch size is 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

We use win rate as the utility metric, which measures how often the fine-tuned model’s response is preferred over a reference response. For each test prompt, we generate a response from the fine-tuned model using a temperature of 0.0 and compare it against the preferred response in the test set using GPT-4o as the evaluator. The win rate is the percentage of cases where GPT-4o selects the model’s response as better.

We adopt win rate because it follows the standard evaluation protocol in recent preference alignment research (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), where a strong LLM is used as an automated judge to estimate human preference judgments. Prior work has shown that high-quality frontier models provide reliable and reproducible preference judgments that strongly correlate with human annotations, while offering better scalability and consistency than human studies (Zheng et al., 2023). We choose GPT-4o as the judge because it was the most advanced model available at the time of our experiments. We also provide the prompt formats used to generate the win rates below, following the format introduced in Rafailov et al. (2024).

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format: Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. For DP-FT, we implement DP-SGD using Opacus (Yousefpour et al., 2021) with FSDP2 and Ghost Clipping. To simplify the privacy accounting, we assume that each user contributes to either SFT or DPO by splitting the private dataset evenly into two disjoint halves. Privacy loss is tracked using the PRV accountant. We perform a grid search over clipping norms $\{0.05, 0.1, 0.5, 1.0\}$ and use 0.1 in our final experiments. Learning rates are searched in $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$, with 5×10^{-5} selected for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, while 5×10^{-6} is used for Llama-3-8B. The batch size is set to 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B. We train for 2 epochs in SFT and 3 epochs in DPO.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

Although DP provides a formal worst-case privacy guarantee, empirical attacks remain informative for understanding how privacy protection manifests in trained downstream models. We therefore perform membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO-fine-tuned downstream models on OpenAssistant and report AUC in Table 5.

Following prior work (Feng et al., 2025), the attack targets preference data for LLM alignment and asks whether a private tuple (x, a^+, a^-) was used during preference alignment. We randomly sample 1,000 training examples as members and 1,000 held-out test examples as non-members. DP-FT does not sufficiently mitigate these MIAs, whereas DPPrefSyn consistently yields lower AUC scores close to random guessing. This result indicates stronger empirical resistance to membership inference, plausibly because downstream fine-tuning uses synthetic preference data rather than real private examples, thereby reducing overfitting to individual private records.

B.5. Ablation Studies on Hyperparameters

This section studies how DPPrefSyn responds to key modeling and privacy-accounting choices: the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filtering threshold, and allocation of the privacy budget. All ablations use OpenAssistant with $\epsilon = 4$.

Varying the embedding model ψ . We evaluate three sentence encoders: BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). Table 8 shows that embedding quality affects absolute performance, but DPPrefSyn remains effective across all three embedding models.

Table 8. DPPrefSyn performance with different embedding models ψ on OpenAssistant at $\epsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . Table 9 compares $p = 10, 20, 30$. The best overall post-DPO performance occurs at $p = 20$, which provides a favorable balance between retaining preference-relevant information and reducing dimensionality for private learning.

Table 9. DPPrefSyn performance under varying DP-PCA dimension p on OpenAssistant at $\epsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . Table 10 evaluates $L = 3, 5, 8$. Increasing L from 3 to 5 improves performance, while $L = 8$ provides no additional gain in this setting, making $L = 5$ a strong default.

Table 10. DPPrefSyn performance under varying number of candidates per prompt L on OpenAssistant at $\epsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying filter threshold. Table 11 compares thresholds 0, 0.5, and 1. A moderate threshold of 0.5 slightly improves over no filtering by removing weak or uninformative preference pairs. In contrast, threshold 1 is too aggressive, reducing data coverage and lowering performance.

Table 11. DPPrefSyn performance under varying filtering thresholds on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying privacy budget allocation. We evaluate alternative splits between DP-PCA (ε_0) and DP-KMeans (ε_1). The default allocation under total privacy budget $\varepsilon = 4$ is $\varepsilon_0 = \varepsilon_1 = 0.5$. We also test two OpenAssistant allocations: more budget to DP-PCA ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$) and more budget to DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$). Table 12 shows that DPPrefSyn is robust to these reasonable allocation changes.

Table 12. DPPrefSyn performance under varying privacy-budget allocations on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

This section examines the use of linear reward models in DPPrefSyn (Line 9). The design follows the Bradley–Terry model with a linear reward function, a common assumption in the preference alignment literature (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023), and is formalized in Section 3.

To assess this modeling choice, we compare the linear model with a fully fine-tuned GPT-2 reward model trained on Anthropic-HH. For each test triplet (prompt, preferred response, dispreferred response), both responses concatenated with the prompt are scored by the trained reward model. A prediction is correct when the preferred response receives the larger scalar reward, and accuracy is the fraction of correctly ranked triplets. The linear model achieves 66% accuracy, while the fully fine-tuned GPT-2 model achieves 62%. This comparison supports the use of the linear reward model: despite its simplicity, it captures preference signal competitively in this evaluation.

B.7. Ablation Studies on Prompt and Response Generator Quality

This section separates the contribution of DPPrefSyn from two possible confounders: public prompt quality and candidate response quality. All experiments use OpenAssistant with $\varepsilon = 4$.

Prompt Quality. To test whether DPPrefSyn’s gains arise from higher-quality public prompts, we rerun DPPrefSyn using OpenAssistant prompts as the public prompt set instead of Alpaca. The DP-FT baseline continues to use the original private OpenAssistant dataset, so both methods operate with *the same set of prompts* from OpenAssistant. Table 13 reports the results.

Table 13. Prompt-quality ablation on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

The results show that DPPrefSyn continues to outperform DP-FT even when both methods use OpenAssistant prompts, indicating that the gains are not solely attributable to higher-quality public prompts.

Candidate Response Quality. To test whether the gains arise from stronger response generation, we construct an enhanced DP-FT baseline. This baseline uses *the same LLaMA-7B-Chat generator* as DPPrefSyn to produce candidate responses for each prompt, ranks the candidates with a trained reward model to create synthetic preference pairs, and then applies DP-FT to the strengthened synthetic preference dataset. This control gives DP-FT responses of comparable generation quality to those used by DPPrefSyn. Table 14 reports the comparison.

Table 14. Candidate-response-quality ablation on OpenAssistant at $\epsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Although the enhanced DP-FT baseline improves over the original DP-FT baseline, DPPrefSyn still achieves higher performance. The advantage therefore cannot be explained solely by access to a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We also conduct a direct head-to-head comparison in which GPT-4o judges DPPrefSyn outputs against DP-FT outputs on OpenAssistant. Table 15 reports the win rate of DPPrefSyn over DP-FT.

DPPrefSyn is preferred over DP-FT in more than 70% of cases across privacy budgets, corroborating the advantage observed in the main evaluation.

Table 15. DPPrefSyn win rate over DP-FT on OpenAssistant.

ϵ	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

This section assesses whether DPPrefSyn clusters correspond to meaningful preference groups. The datasets used in the main experiments do not include explicit human annotator group information, making it difficult to assign a ground-truth preference style to each DP-KMeans cluster. Following Chakraborty et al. (2024), we therefore simulate two annotator groups with distinct preferences using GPT-4. For each Alpaca prompt (Taori et al., 2023), GPT-4 generates two responses under different instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with the two generated responses, a simulated preference label, and an annotator type (“elementary school student” or “PhD student”). We then apply DP-KMeans with $\epsilon = \infty$ to the embedding differences of the paired responses. The resulting clusters strongly align with the simulated groups: one cluster is dominated by PhD student preferences (95.6%), and the other is dominated by elementary school student preferences (93.1%). This result supports the interpretation that DPPrefSyn clusters can encode meaningful preference styles.

B.10. Samples from DPPrefSyn

In the section, we present generated samples using DPPrefSyn. Representative examples for each task are provided in Table 16. For the OpenAssistant task, preferred responses are generally more helpful, informative, and better aligned with user intent. In the spelling correction example, the preferred response explains each error, while the less preferred

Table 16. Generated samples from DPPrefSyn with $\epsilon = 4$. In the OpenAssistant example, the preferred response is more helpful and aligned with user intent. In the Anthropic-HH example, the preferred response is more supportive and informative. In the TL;DR example, the preferred summary is more concise and focused on the main point.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

response simply states the number of mistakes. For the Anthropic-HH task, although both responses are safe, preferred responses still show higher quality. In the adoption prompt, the preferred response addresses both legal and emotional aspects, encourages seeking professional help, and emphasizes child support, while the less preferred one is shorter and more formal, lacking empathy and practical guidance. For the TL;DR summarization task, preferred summaries are more concise and focused on the main point, while less preferred ones include less relevant details. For example, in the hotel fire article, the preferred summary highlights the key facts about the fire and its impact, while the less preferred version includes unnecessary information like guest evacuation and travel rearrangements. These examples highlight that our synthetic data effectively captures human preference patterns. We provide more samples from DPPrefSyn below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\epsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4.

Therefore, $4/16$ is equivalent to $1/4$!

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* "Finnished" should be "finished"

* "resturant" should be "restaurant"

I hope this helps! Let me know if you have any other sentences you'd like me to check.

Less Preferred Response: The sentence “He finnished his meal and left the resturant” contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night’s sleep.
3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.
7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one’s way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent
2. Sleeping bag
3. Sleeping pad
4. Backpack
5. Flashlight or headlamp
6. Water bottle or hydration bladder
7. Portable stove or camping grill
8. Cooking pot and utensils
9. Toiletries (toothbrush, toothpaste, wipes, etc.)
10. First aid kit

Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I’m here to help you classify the objects you’ve listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you’ve listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I’d be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I’m here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\varepsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I’m so sorry to hear that you’re considering giving your kids up for adoption. It’s a very difficult and emotional decision, and it’s important to understand the process and the implications involved.

Firstly, it’s important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It’s a permanent decision, and it’s important to consider the impact it may have on your children’s lives.

If you’re considering adoption, it’s important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It’s also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It’s important to be honest and open with them about what’s happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\epsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been “no hostage crises”, adding that the man was “alone in the room”.

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

“His firearm was taken away,” Dr Kutlubay said, adding that the gun in the officer’s possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year’s Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer’s possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the “investigation is now being conducted by the IPCC”.

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It’s the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.